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Section: A2

Department: EEE

Group Name: WaveSync

Group members:

Farhan Tanvir	230021112
Nadman Ibne Mamun	230021114
Goolam Muktadir Hameem	230021144

Pressure Performance

Statistical and Stochastic Analysis of NBA Free-Throw Performance Under Clutch Pressure

NBA Regular Season & Playoff Data, 2006–07 to 2015–16

Abstract

This project investigates whether NBA players perform differently at the free-throw line in clutch situations, used here as an observational proxy for pressure. Each free throw is modeled as a Bernoulli random variable (make = 1, miss = 0), allowing estimation of success probability, mean, variance, and confidence intervals. The analysis combines pooled two-proportion z-tests, paired player-season and career-level tests, a 20,000-resample player-level bootstrap, individual-player comparisons, and sample-size correlation analysis. **Because the supplied dataset is aggregated rather than chronological shot-by-shot data, consecutive-attempt correlation cannot be estimated and is explicitly treated as a limitation rather than fabricated.**

1. Research Question & Hypotheses

Research question: Do NBA players' free-throw success probabilities change in clutch situations, and is any apparent effect stable across athletes rather than an artifact of limited clutch sample sizes?

- Primary athlete-level null hypothesis: $\text{mean}(\text{Clutch FT\%} - \text{Non-Clutch FT\%}) = 0$.
- Directional “choke” alternative: $\text{mean}(\text{Clutch FT\%} - \text{Non-Clutch FT\%}) < 0$.
- Attempt-level Bernoulli null hypothesis for the two-proportion z-test: $p_{\text{clutch}} = p_{\text{non-clutch}}$.
- Attempt-level directional alternative: $p_{\text{clutch}} < p_{\text{non-clutch}}$.

Important: the data are observational. “Clutch” is an operational proxy for pressure, so the analysis supports associations, not a causal claim that psychological pressure itself caused a change in performance.

2. Data & Methodology

2.1 Dataset and data correction

- 461 player-season records covering 200 unique players from 2006–07 through 2015–16.
- The spreadsheet reports total attempts and FT%, plus clutch attempts and clutch FT%, separately for the regular season and playoffs.
- Nineteen player-season rows have zero playoff clutch attempts with clutch FT% stored as 0; these are treated as missing clutch observations, not 0% shooting performances.
- Season-total FT% already includes clutch attempts. Exact integer makes are recovered and clutch makes/attempts are subtracted from the total to create a genuinely non-overlapping non-clutch baseline.

2.2 Bernoulli and stochastic model

For a free throw in condition c , define:

$X_{c[n]} = 1$ for a made free throw; $X_{c[n]} = 0$ for a miss

$$P(X_{c[n]} = 1) = p_c, \quad E[X_{c[n]}] = p_c, \quad \text{Var}(X_{c[n]}) = p_c(1 - p_c)$$

If the attempts were an IID Bernoulli process, non-zero-lag covariance and correlation would be zero. However, the supplied file contains only aggregate makes/attempts and percentages, not chronological shot order. Therefore the IID assumption can be stated as a model, but lag-1 or consecutive-shot correlation cannot be empirically estimated from this dataset.

2.3 Statistical procedures

- Wilson 95% confidence intervals for Bernoulli success probabilities.
- Two-proportion z-tests using pooled clutch and non-clutch makes/attempts (attempt-weighted analysis).
- Paired t-tests and one-sided “choke” tests on player-season FT% differences, with Wilcoxon signed-rank checks and Cohen’s dz.
- Career-level aggregation so each eligible player contributes one regular-season career difference; a minimum of 10 career clutch attempts is used for the player-level analysis.
- Player-level nonparametric bootstrap with 20,000 resamples to validate the mean career clutch effect without relying only on the t-test.
- Spearman correlation between career clutch-attempt volume and the signed/magnitude clutch effect to assess small-sample instability.

3. Results

3.1 Bernoulli success probability, mean, and variance

Condition	Makes	Attempts	$p = E[X]$	$\text{Var}(X)$	95% Wilson CI
Regular Non-Clutch	92,862	120,353	0.7716	0.1762	[0.7692, 0.7739]
Regular Clutch	20,105	25,815	0.7788	0.1723	[0.7737, 0.7838]
Playoff Non-Clutch	18,147	23,862	0.7605	0.1821	[0.7550, 0.7659]
Playoff Clutch	4,089	5,244	0.7797	0.1717	[0.7683, 0.7908]

Key point: When every attempt is pooled, regular-season clutch success is 77.88% compared with 77.16% non-clutch. This is an attempt-weighted comparison, not an equal-player comparison.

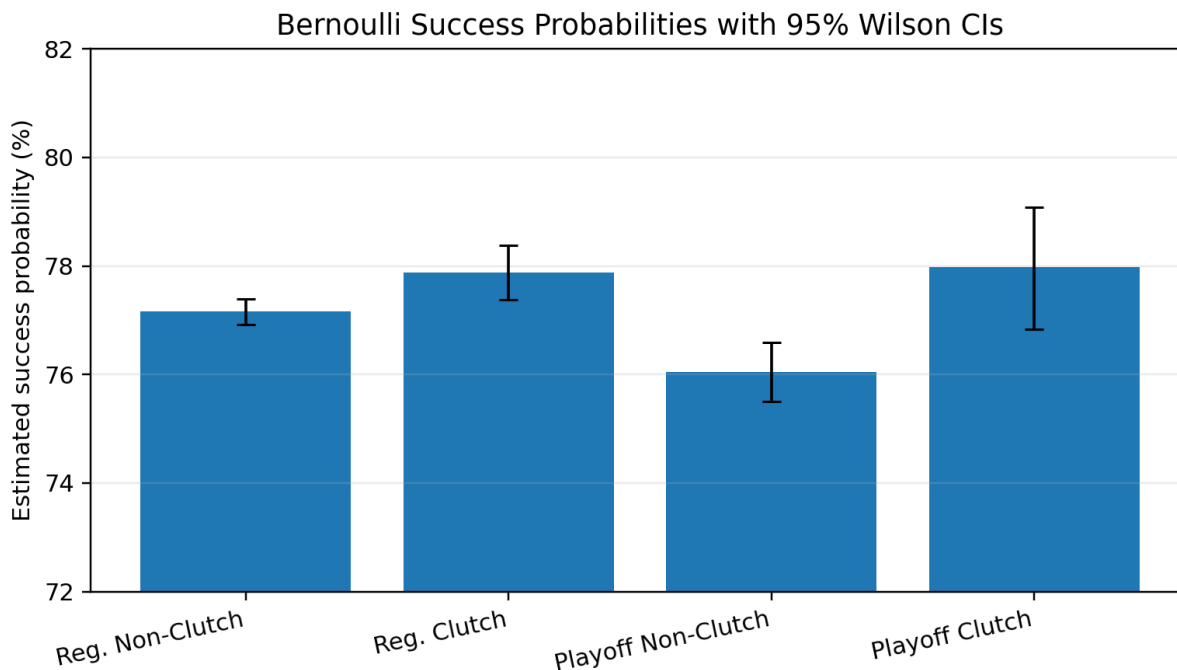


Figure 1. Pooled Bernoulli success probabilities with 95% Wilson confidence intervals.

3.2 Two-proportion z-test: attempt-weighted result

Analysis	Clutch – Non-Clutch	z	p (2-tailed)	p (choke: < 0)
Regular season	+0.72 pp	2.516	0.0119	0.9941
Playoffs	+1.93 pp	2.972	0.0030	0.9985

Interpretation: The pooled regular-season test gives a small increase of +0.72 pp, not a decrease ($z = 2.52$, two-tailed $p = 0.0119$). The directional choking alternative is therefore not supported by the pooled attempt-level test. The playoff pooled result is also positive. These pooled tests heavily weight players with many attempts and do not account for clustering by player.

3.3 Equal-player/player-season comparison

Analysis	n	Mean diff	t statistic	p (2-tail)	p (choke)	Cohen dz
Regular: all player-seasons	461	-1.00 pp	$t(460) = -2.62$	0.0090	0.0045	-0.122
Regular: ≥ 10 clutch attempts	448	-1.10 pp	$t(447) = -2.94$	0.0034	0.0017	-0.139
Playoffs: all player-seasons	442	+1.17 pp	$t(441) = 1.31$	0.1918	0.9041	0.062
Career player-level (≥ 10)	194	-1.27 pp	$t(193) = -2.34$	0.0203	0.0102	-0.168

Regular-season finding: With equal player-season weighting, clutch performance is lower by about 1 percentage point. The all-row paired result is -1.00 pp ($p = 0.0090$), and the ≥ 10 -attempt subset is -1.10 pp ($p = 0.0034$). Effect sizes are small.

Playoff robustness warning: The playoff paired t-test is not significant ($p = 0.1918$), while the Wilcoxon signed-rank test is significant ($p = 0.0338$). Therefore the playoff tests are not consistent in significance; the playoff result should be described as unstable/noisy rather than as a confirmed effect.

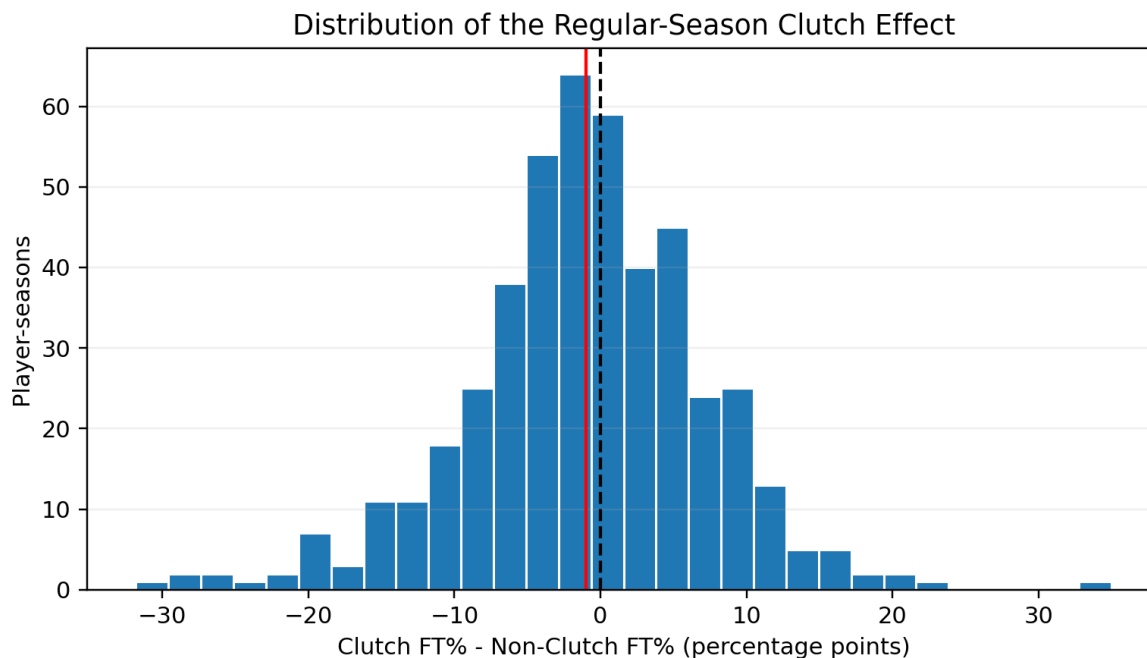


Figure 2. Distribution of regular-season player-season clutch effects. The average is slightly below zero.

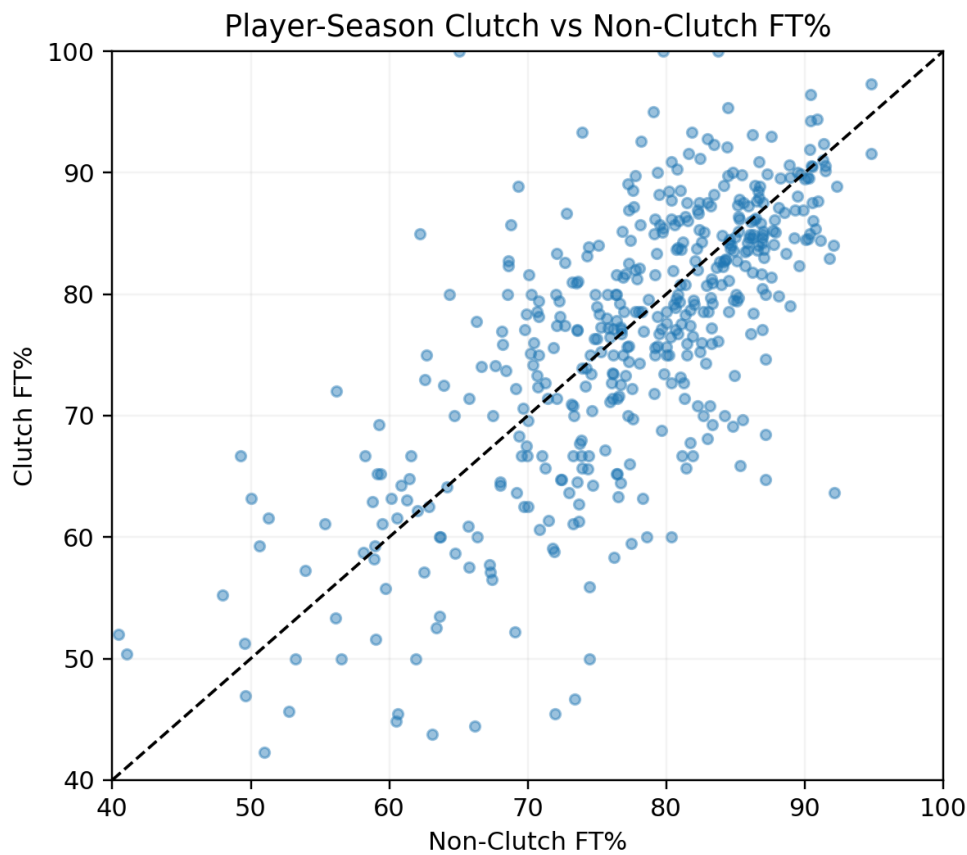


Figure 3. Clutch versus non-clutch FT% by player-season. Points below the diagonal are worse in clutch situations.

3.4 Career player-level bootstrap validation

Bootstrap setup: The 194 eligible players were resampled with replacement 20,000 times. For each resample, the mean career difference (Clutch FT% – Non-Clutch FT%) was recomputed.

Bootstrap statistic	Result
Observed career mean difference	-1.27 pp
Bootstrap mean	-1.26 pp
Bootstrap 95% percentile CI	[-2.32, -0.20] pp
One-sided bootstrap p (choke)	0.0088

Interpretation: The bootstrap 95% CI (-2.32 to -0.20 pp) lies below zero, supporting the small negative athlete-level association found by the paired t-test. This validates the regular-season player-level conclusion without depending solely on a normality assumption.

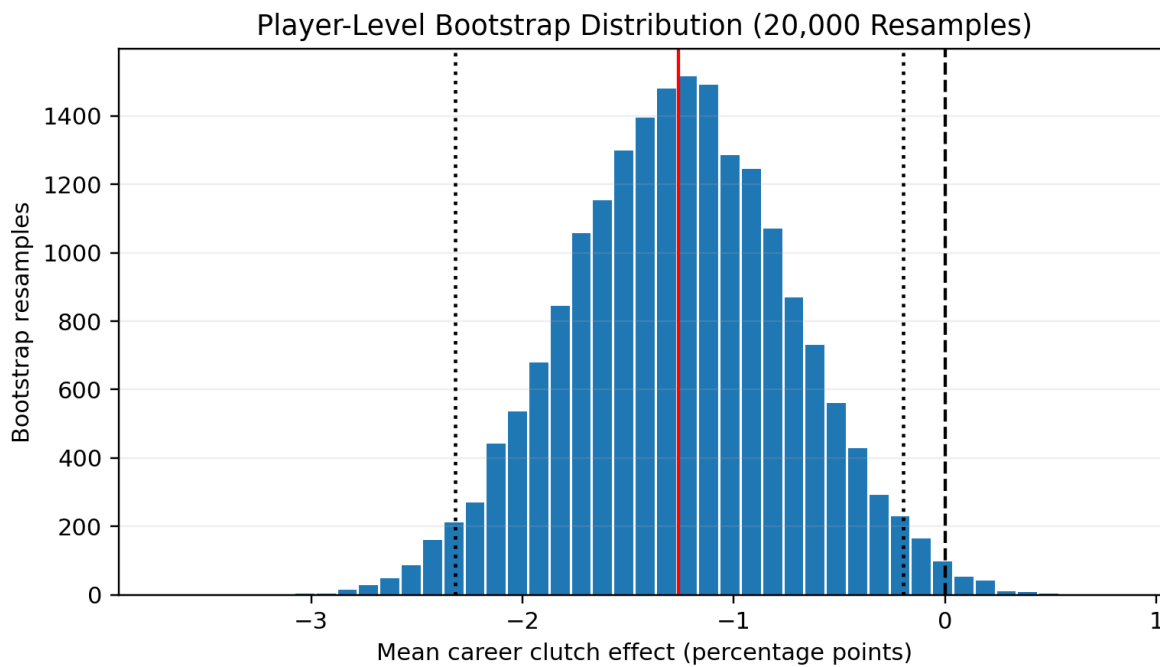


Figure 4. Bootstrap distribution of the mean career clutch effect. Dotted lines mark the 95% percentile interval; dashed line marks zero.

3.5 Why the pooled and player-level answers differ

The two analyses estimate different quantities. The pooled z-test weights every shot equally, so high-volume shooters contribute many more observations. The paired career analysis weights each eligible player equally after forming that player’s career clutch and non-clutch rates. In this dataset, the pooled regular-season rate is slightly higher in clutch, while the average player-level difference is lower. This is a weighting/composition effect similar to Simpson’s paradox and is not a computational contradiction.

For the project question “do athletes perform worse under pressure?”, the player-level paired estimate and player bootstrap are more directly aligned with the average-athlete interpretation. The pooled Bernoulli result remains important because it describes the success probability of an attempt drawn from the full observed shot pool.

3.6 Individual differences and small-sample noise

The magnitude of a player’s clutch deviation decreases strongly as career clutch-attempt volume grows:

Spearman $\rho = -0.441$, $p = 1.21e-10$. The signed correlation is weak and non-significant, so more attempts do not systematically push the effect positive or negative; they mainly reduce extreme deviations.

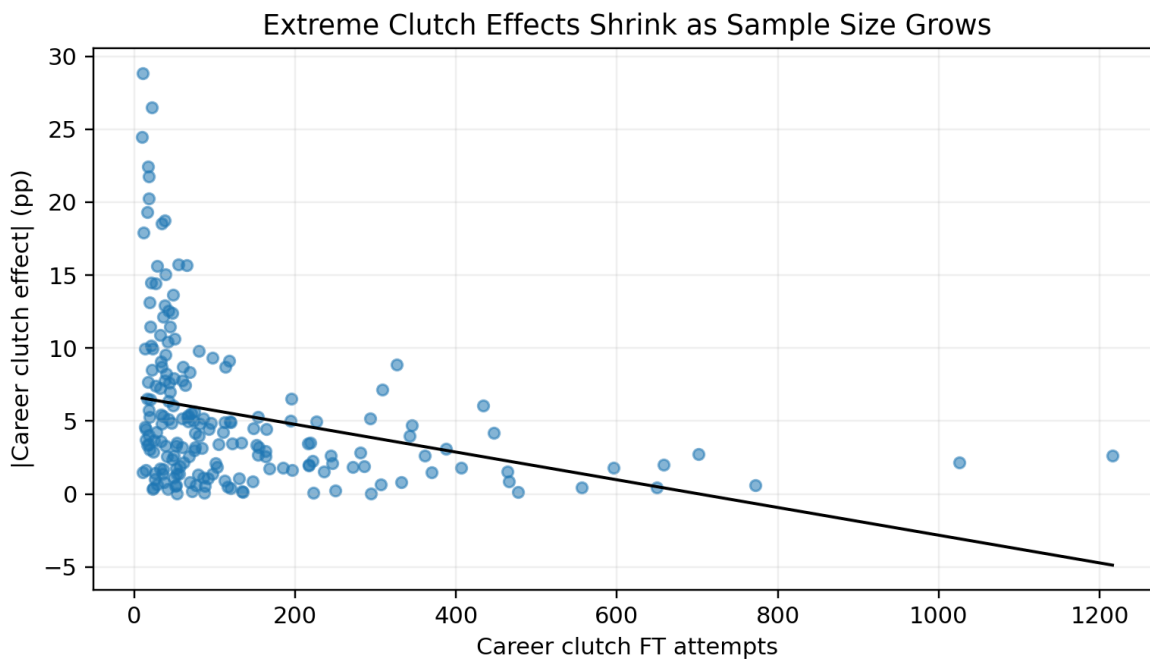


Figure 5. Absolute career clutch effect versus career clutch attempts. Extreme estimates are concentrated at small sample sizes.

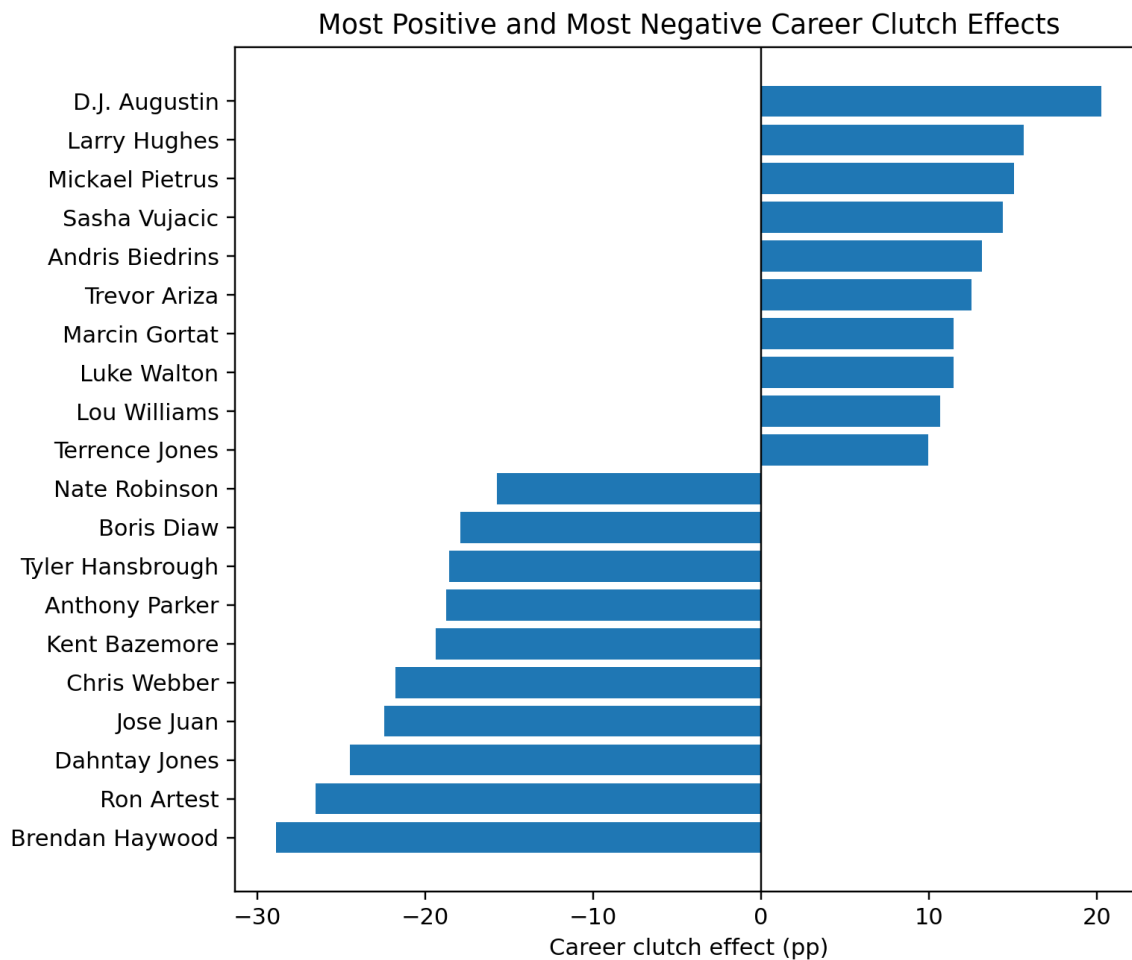


Figure 6. Most positive and most negative career clutch effects among players with at least 10 career clutch attempts. Rankings should not be interpreted as stable traits because many extremes have small clutch samples.

3.7 Season-to-season pattern

The mean player-season clutch effect varies substantially by season. Most seasons are near zero, while 2010–11 and 2012–13 show larger negative averages. This instability argues against a single fixed “choke” effect of the same magnitude in every season.

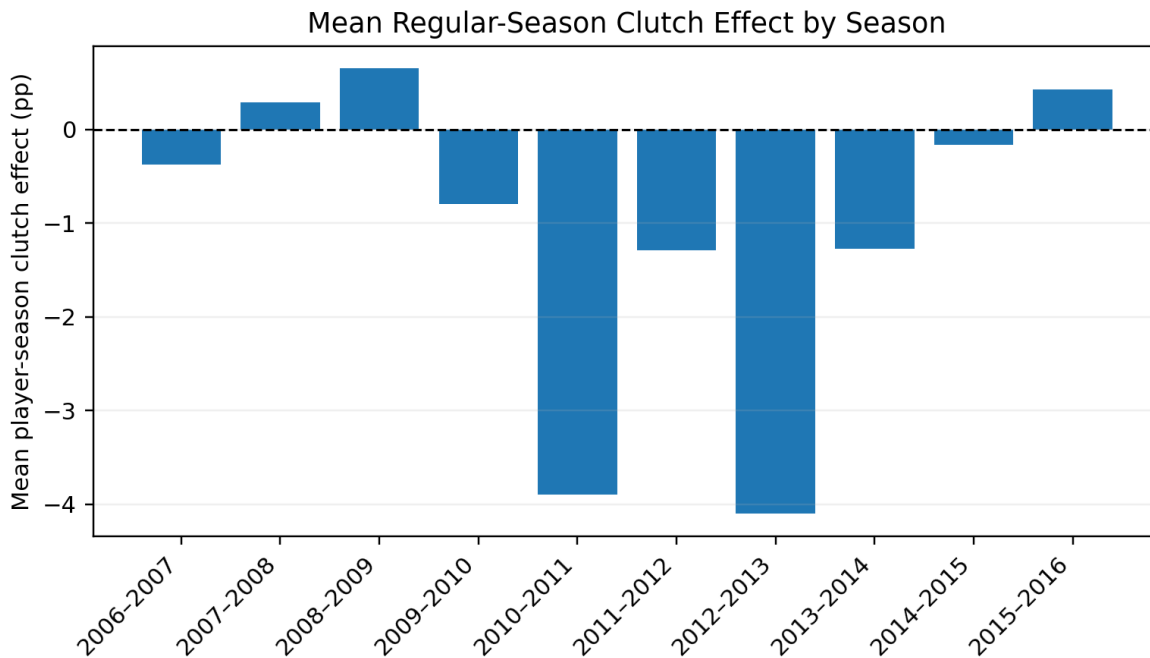


Figure 7. Mean regular-season player-season clutch effect by season.

3.8 Consecutive-attempt correlation: what can and cannot be concluded

Not estimable from the supplied data. A correlation such as $\text{Corr}(X[n], X[n+1])$ requires the order of individual makes and misses. The spreadsheet contains only aggregate counts and percentages, so it is impossible to identify which attempt followed which. Any numerical lag-1 correlation generated from these aggregate data would be artificial.

- What is supported: Bernoulli success probabilities, means, variances, confidence intervals, pooled proportion tests, paired player effects, bootstrap uncertainty, and cross-player/sample-size correlations.
- What is not supported: empirical serial autocorrelation, hot-hand/cold-hand dependence, run-length distributions, Markov transition probabilities, or other chronological sequence statistics.
- Future work: obtain play-by-play shot-level data with timestamps and clutch labels; then compute lagged covariance/correlation and compare dependence patterns between clutch and non-clutch conditions.

4. Final Statistical Summary

Analysis	Weighting	Difference	p-value	Conclusion
Pooled regular Bernoulli	Attempts	+0.72 pp	0.0119	Clutch slightly higher when shots are pooled
Regular player-season	Equal player-season	-1.00 pp	0.0090	Small significant drop
Regular ≥ 10 clutch att.	Equal player-season	-1.10 pp	0.0034	Small significant drop

Career player-level	Equal players	-1.27 pp	0.0203	Small significant drop
Career bootstrap	Equal players	-1.26 pp	0.0088	95% CI below zero
Playoff paired t-test	Equal player-season	+1.17 pp	0.1918	Not significant; robustness tests disagree

5. Key Findings & Conclusion

- At the attempt-pooled Bernoulli level, regular-season clutch success is 77.88% versus 77.16% non-clutch, so the pooled z-test does not support choking.
- At the player level, the average career clutch effect among 194 players with at least 10 clutch attempts is -1.27 pp, with paired $p = 0.0203$ and bootstrap 95% CI [-2.32, -0.20] pp. This supports a small negative association for the average player.
- The practical effect is small (career Cohen's $d \approx -0.168$), so the data do not support a dramatic universal collapse in ability under clutch pressure.
- Individual clutch/choke extremes become much smaller with more clutch attempts ($\rho = -0.441$, $p < 0.0001$), indicating substantial small-sample noise in player rankings.
- Playoff evidence is unstable: the paired t-test and Wilcoxon test disagree in significance, so a strong playoff conclusion is not justified.
- The supplied aggregate dataset cannot test consecutive-shot correlation. A shot-by-shot dataset is required for that part of a full stochastic-process analysis.

Overall conclusion: The strongest defensible conclusion is that clutch situations are associated with a small decline in the average player's free-throw percentage, while the shot-pooled league-wide probability is not lower. The apparent individual "clutch" and "choke" labels are highly sensitive to sample size. Because this is observational aggregate data, the project should not claim a large universal or causal pressure effect.

6. Limitations & Future Work

- The dataset's definition of "clutch" is inherited from the supplied source file and was not re-derived from raw play-by-play.
- Non-clutch performance is reconstructed arithmetically from total and clutch makes/attempts; it therefore inherits any reporting or rounding conventions in the supplied data.
- Repeated player-seasons are not independent at the raw row level; the career-level analysis and player bootstrap reduce this problem, but a multilevel model would be even stronger.
- Many players still have modest clutch sample sizes, limiting reliable player-specific conclusions.
- The pooled z-test assumes independent Bernoulli attempts and ignores within-player clustering; it is included because it is part of the planned Bernoulli analysis but should not be treated as the sole inferential result.
- Future work should use shot-level play-by-play data and a hierarchical/Bayesian model. Shot-level order would allow empirical autocorrelation, transition probabilities, streak analysis, and a direct stochastic-process comparison under clutch and non-clutch conditions.

7. Reproducibility & Deliverables

The final MATLAB script supplied with this report performs the complete corrected analysis from the original Excel dataset and writes reproducible outputs to a final_outputs folder. It includes:

- Bernoulli probability/variance/Wilson CI summary CSV
- Two-proportion z-tests
- Paired t-tests, directional tests, Wilcoxon checks, and effect sizes
- Career aggregation and 20,000-resample player bootstrap
- Player rankings and sample-size correlations

- Season trend output
- Seven analysis figures and summary CSV files

8. Data Source & Method References

1. Project dataset: Starter_FT_Stats_2006_2016_2.xlsx (supplied dataset; 461 player-season rows, 2006–07 to 2015–16).
2. Final analysis code: Pressure_Performance_Final.m (supplied with this report).
3. Efron, B., & Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall/CRC.
4. Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158), 209–212.